Omnichannel ERP (Enterprise Resource Planning) Integration Platform for Retail, Inventory, and Shopee Synchronization

(Cloud Solution Proposal / Cloud Application Design)

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INTRODUCTION

Background of the Study

In the contemporary retail landscape, hybrid sales models aren't just a trend, but a necessity for businesses primarily to attend to the consumers' insatiable desire for seamless services. Evidently, large franchises and enterprises have already been utilizing this approach for years, and the imminent adaptation of small businesses is not a stretch. A physical store alongside digital storefronts opens a lot of sales opportunities, making it not just an ideal setup but a must in order to survive in the brutal and competitive electronic commerce ecosystem. And with these innovations in modern entrepreneurship, online platforms for both selling and buying emanated (Lazada, Shopee, Amazon, etc.). Online marketplaces such as Shopee paved a way for much more accessible business-to-consumer (B2C) and consumer-to-consumer (C2C) platforms, removing the heavy lifting so traders won't even have to start from scratch of development, but a plug and play machine was already laid out, both merchants and consumers can start from the get go.

While digital channels for sales are the leading trend, physical stores can't be eliminated, as they are still relevant. A non-negligible number of consumers still prefer going to physical stores where they can examine and decide in real

time, and implementing multi-channel retailing still encounters discrepancies and challenges (Chen, J. et al, 2022). Traditionally, physical point of sale (POS) systems and inventory databases operate independently, as well as online stores, and these isolations introduce several technical challenges. In order to resolve these problems, advanced centralized infrastructure must be implemented to act as a bridge across multiple channels. As highlighted by Febriani et al. (2025), modern tools and cloud-based systems are the key in overcoming the systemic barriers in the SMEs. The impact of cloud-based Enterprise Performance Management (EPM) was closely investigated by Raghu Praneeth Akula in his study in 2024 titled *The Impact of Cloud-Based Enterprise Performance Management on Inventory Optimization and Financial Agility in Omnichannel Retail*. In conclusion of the study, improvement across all boards was conspicuous, inventory and data integration among others was ameliorated. Imperative successes made in this study were vital in every modern system, namely, organizational data governance, collaboration frameworks and change management.

This project proposes the design of a cloud-based Omnichannel Enterprise Resource Planning (ERP) platform. The system is engineered to act as a centralized hub connecting a retailer's inventory, physical store and the Shopee Open API. Synchronization multi-directionally is the main goal, from the point of stocking the inventory, the physical store's sales should automatically be updated, in restock or sales, and these changes should be reflected in all the main channels. High efficiency, responsiveness and performance is guaranteed

utilizing Huawei Cloud technologies, as their services are committed to dynamic capabilities to meet the e-commerce demands.

PROBLEM STATEMENT

Modern businesses usually have their online side of retailing, and in the nature of digital commerce, operational sales tracking is a requirement. Considering inventory and online listing synchronization, in addition to physical storefronts or offline-channel, discrepancies and data misalignment are common. Most of these businesses rarely have an omnichannel synchronization as to if these enterprises have any database for any of the sales mode and inventory, likely are isolated, no central system which leads to dissociated channels.

Small and medium sized enterprises are the primary target users of this project. As they are expected to operate utilizing modern programs, but are often isolated and run independently. Synchronized master inventory to both in-store point-of-sale (POS) and digital marketplace or Shopee Open API is the main goal of this project. Citing Akula, R. P. (2024), inventory optimization omnichannel retail, improvement in the entire revenue operation is perceptible. The Omnichannel-ERP project sought to augment the limited capability of SMEs' ability to connect inventory to their physical store and online market.

SYSTEM REQUIREMENTS

Functional Requirement

- **Real-Time Inventory Synchronization**

  Multi-directional stock updates with master inventory at the center, and between the physical Point-of-Sale (POS) and the Shopee Open API.

- **Order Lifecycle Management**

  Unified tracking of orders from checkout to packing and dispatch. This is the overview of the ideal working omnichannel-erp where it promotes automation to avoid human error and data misalignment encountered by isolated systems in the traditional sales models.

- **Centralized Product Catalog**

  Master inventory database for products, variations, and pricing.

- **Automated Low-Stock Alerts**

  System-triggered automation in the main dashboard, where it displays the exact product that is critically low in number. This is to oblige the workforce to restock those products.

- **Inventory Restocking**

  With the central database system and automated low stock alerts, the restocking process is seamless, since a dedicated adding and confirming receipt is laid out.

- **Shopee Product Mapping (Linking)**

Binding local inventory SKUs directly to active Shopee storefront listings, creating a bridge for accurate data synchronization. This is the key for inventory, physical point of sale (POS) and the Shopee products concurrence.

- **Sales Analytics, History and Revenue Reporting**

Dashboards tracking revenue, top-selling SKUs, and the entire sales history, revenue and profit reporting are exhibited in the main database system and these dashboards are only available to the admin view.

- **Role-Based Access Control (RBAC)**

User permissions to secure specific modules for monitoring staff, cashiers, clerks, and administrators. This is the initial security feature so that no intruder can access any sensitive information and detail of the system.

- **Automated Document Generation**

Generation of PDF waybills and pick-tickets for online orders. Pick tickets are guides for the packaging team.

Non-Functional Requirements

- **Scalability**

The system is designed to handle fluctuating workloads, particularly in peak sale events of Shopee, having the ability to operate in times where orders reach hundreds within a minute.

- **Availability**

The system's uptime is a crucial matter for a business, making products online leave it for consumers to order items in their desired time. Application can't stop, anytime, sales may register to the system and if the system is not up and running, large loss of sales is imminent. That is why this project is designed to handle the volatility of the number of orders.

- **Security**

Previously defined Role-Based Access Control (RBAC) is strictly enforced to ensure only authorized personnel can access or modify critical data. Business metrics exhibited in the dashboard mentioned, Shopee API credentials and customers' personal information are among the most sensitive data that can't be exposed, and need rigorous encryption to avoid information leakage.

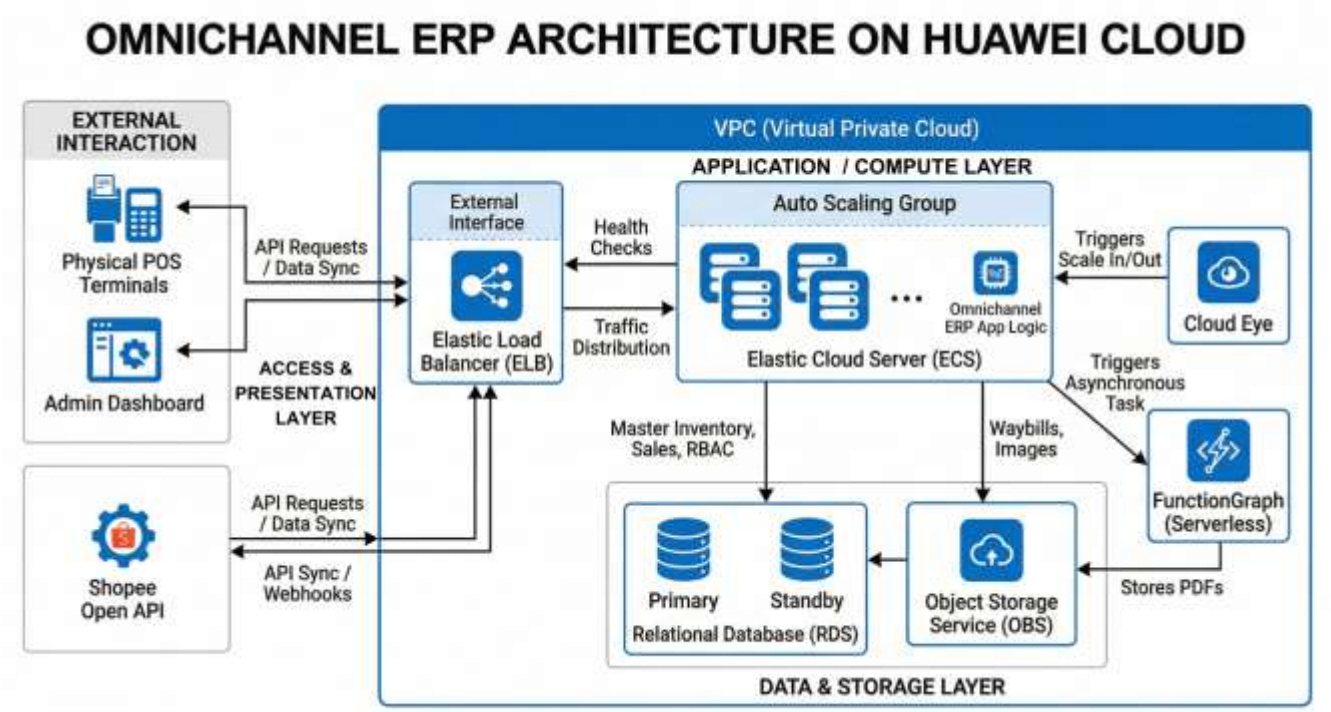
- **Performance**

Omnichannel ERP is designed to perform on a high level to attend to the requirement of minimal stutter in the system and to perform in a continuous retail process across all sales channels. Latency poses a threat to

the main system data alignment, to mitigate, the program must execute in sub-second response times.

PROPOSED CLOUD ARCHITECTURE

Architecture diagram



Access / Presentation Layer

This layer handles all incoming external traffic, HTTP/HTTPS requests from physical POS terminals (in-store cashiers), the admin web dashboard, and webhooks transmitted by the Shopee Open API.

Application / Compute Layer

This layer is the core engine of the Omnichannel-ERP. It contains the application logic responsible for inventory deductions, Shopee product mapping,

role-based authentication, and other background processes that make the operation run smoothly and coherently.

Data / Storage Layer:

The foundation of the system where the master inventory, sales history, and generated files reside. It consists of a managed relational database for structured data and scalable object storage for unstructured data (PDFs and images).

Interaction between components

When an online purchase occurs, the Shopee Open API sends data in real time to the system via HTTP post. That is the initiation of the entire process of automated sales lifecycle management using Huawei Cloud technologies. In every external payload, it is the Elastic Load Balancer (ELB) that receives. And ELB as its main responsibility is to distribute traffic, it then routes that data from external interaction such as a Shopee purchase to the healthiest Elastic Cloud Server (ECS) instance. Housing the compute layer of the system, it then processes the order instantly deducting the stock from the master inventory for unified omnichannel data, alongside revenue and other logic that's in the application and compute layer. Simultaneously, the same layer triggers a serverless function via FunctionGraph that generates the PDF waybill and pick-ticket. Then those generated files are automatically saved into the Object Storage

Service (OBS). Even if these processes look like they are working independently, that is how cloud technology is designed for interacting modules which ensures that the main omnichannel ERP system is highly responsive.

CLOUD SERVICE SELECTION

- **Virtual Private Cloud (VPC)**

  The Omnichannel-ERP program is designed for businesses, primarily SMEs, and since it involves transaction of monetary processes, security is the top priority and VPC is the best option among the rest. IT places the main database and application servers within private subnets to protect from direct public internet vulnerability. Preventing cyber attacks that can cause catastrophic damage to the business and the information of the consumers.

- **Elastic Load Balancer (ELB)**

  It enables automated distribution of traffic within the system handling volatile purchase traffic. ELB's role in ensuring that data processing is not overwhelmed by intelligently managing the payloads that are routed across the entire cloud channels.

- **Elastic Cloud Server (ECS)**

  It is safe to say that Elastic Cloud Server is the brain of the entire system. Reliability of its computing power is the selling point of the service. It hosts the ERP logic, and computes the necessary backend operations that enable triggers and guarantee efficient data reflection.

- **Auto Scaling (AS)**

Auto Scaling works in tandem with the ELB and ECS in distributing and terminating at the same time the compute resources based on real-time system demand. This automation ensures the Omnichannel-ERP maintains sub-second response times during severe traffic registers, at the same time optimizing cloud service costs by scaling down during off-peak hours.

- **Object Storage Service (OBS)**

Providing highly scalable and cheaper storage costs, OBS is the best option in storing generated files and documents like PDF waybills and pick-tickets for the packaging team. Offloading these relatively large files from ECS hard drives, capacity exhaustion and server failure are avoided.

- **FunctionGraph**

FunctionGraph is utilized as a serverless engine that executes event driven operations. Processes like waybill PDF generation, automated low-stock alert calculations and pick-ticket generation in the background provide tremendous help to the ECS by taking these loads of operation making it more seamless and automated processes.

SCALABILITY AND AVAILABILITY DESIGN

One of the primary requirements of any online system is to keep it running 99% of the time. And to guarantee that, preventive counter-measures should be placed. Core compute and storage components namely Relational Databases (RDS) and Elastic Cloud Server (ECS) are distributed across Availability Zones which are these extensive physical data centers within Huawei Cloud region. This maintains insulation of the system's operation from any outages, jumping off from halted zones to any secondary healthy one. Securing zero data loss and misalignment ensures no operational disruption.

Health checks were one of the routines of Elastic Load Balancer (ELB) shown in the architecture diagram, inspecting all ECS instances. Explain in the cloud service selection part that ELB acts as the traffic distributor, responsible for receiving payloads from external interaction and routing them to healthy servers. Any unresponsive ECS instance upon ELB health check, it stops forwarding traffic to it, isolating the fault.

Auto Scaling (AS) was configured with alarm features, simultaneous monitoring of CPU. If the threshold of CPU usage is surpassed, automated actions are set to dynamically provision new ECS instances. Likewise, in times of off-peak traffic, unused extra servers are terminated.

SECURITY DESIGN

Discussed in the security section that the system would only be accessed according to the role that would be assigned by the admin. Role-Based Access Control (RBAC) is a security layer utilizing tiered user permission guaranteeing no unauthorized entity or personnel can access and alter sensitive system details. Roles are defined as monitoring staff, cashier, clerks, and the admin, interacting to an interface specified to their role. The same in the infrastructure level, Identity and Access Management (IAM) is utilized so no unwanted intrusion and modification of the Huawei Cloud resources.

Businesses' top priority is security, isolating the entire compute and data storage layer is imperative. Mentioned in the cloud services selection was the Virtual Private Cloud (VPC). Deploying the brain that operates on transactions and database data into private subnets protecting from any internet susceptibility. And that is why ELB was utilized for the distribution of external payloads acting as traffic director, no direct path to important internal layers to shield the system from imminent cyber attacks.

Encryption is strictly enforced, in case of breach, data would not be of any use because of rigorous data encryption protocols. Admin dashboards, Shopee API credentials and customers' personal information are encrypted, alongside data traffic transmitted from in store (POS), Shopee platform and the main cloud using SSL/TLS protocols. Additionally, OBS which stores PDF waybills and pick-tickets are encrypted at rest.

Acting as a virtual version of windows firewall, imposing strict rules to protect the internal layers exclusively. Directing traffic at the server level is the essential responsibility of security groups. Limiting the access of ELB for internal communication, locking the internal database ports to accept queries exclusively from the verified Application Layer, and blocking any outside network access attempts.

COST AND RESOURCE CONSIDERATION

Leveraging a pay-as-you-go service pricing instead of purchasing and maintaining physical servers, SME's expenses are scaled down. Furthermore, integration of Auto Scaling (AS) is economically vital, paying only for the peak of traffic using the extensive compute engine of the system.

The proposed operational architecture was designed and engineered to the most frugal and efficient cloud services available. Automated generation of PDF waybills and pick tickets for example, instead of storing the files to expensive ECS hard drives, they are intelligently offloaded to OBS which offers massive capacity at a fraction of the cost. FunctionGraph's addition to the cloud services was for background automation and alert features, ensuring that the system is only billed for the exact millisecond function executed.

Layers of preventive measures are put in place to maintain the system operational whenever needed, at the same time being efficient financially is a great feat in practicing sustainability. Utilizing Huawei Cloud Services and

architectural design proposed to guarantee a drastic overall energy consumption reduction, perfectly aligning with modern green computing standards.

EXPECTED BENEFITS

At the infrastructure level, integration of Huawei Cloud's Elastic Load Balancer (ELB), Auto Scaling and other services enables the system to automatically distribute the operation and automate scaling of server usage while provisioning extra unused instances to meet the real-time demands. Omnichannel-ERP web application was also designed to achieve certain scalability for it to have a user friendly interface.

Transitioning to a cloud native environment, businesses can eliminate the expensive purchasing and maintenance of physical servers. Selecting Huawei Cloud services that are relatively cheap and efficient to attend to the system's operational requirements. Utilizing pay-as-you-go services, automated scaling, and offloading large redundant files to a high capacity but economical storage service OBS reduces the overall expenditures.

Availability is ensured with the proposed cloud application design. Simultaneous health checks of instances by the ELB and deployment of the core compute and data storage components across multiple availability zones established a fault tolerant architecture. Loss of sales, data loss and misalignment are prevented demonstrating high system reliability.

Well distributed processes, exhibited in the designed architecture, prove the efficiency of the entire system. From automating operations, backend logic processes and data allocation are well orchestrated to eliminate the problems of multiple channel synchronization.

CONCLUSION

The application Omnichannel-ERP proposed an optimal solution for modern SMEs current systems. This project presents a clear plan and design for the process of adapting to the contemporary way of retailing. Utilizing Huawei Cloud technology, the core design of the cloud system was based on the available services. Carefully deciding which actually can be exploited at the maximum level and considering the financial concept in the process. The Omnichannel-ERP cloud architecture design selected services are Virtual Private Cloud (VPC) for rigorous network isolation, paired with Elastic Load Balancer (ELB) handling volatile data traffic, with Auto Scaling (AS) at the background and Elastic Cloud Servers (ECS) as the main data processor. Strategically storing a high capacity yet economical storage service, Object Storage Service (OBS) and tapping on the capabilities of FunctionGraph for background automation achieving a highly available and scalable system.

This architecture will fundamentally upgrade any enterprise operations by bridging the gap between master inventory, physical storefronts and Shopee store counterpart. Accurate and responsive omnichannel synchronization across these important sections which are traditionally isolated is the main selling point of the system, eliminating problems of loss of sales, data misalignment, overselling and minimizing human error brought by undefined user distinction and interface flaws.

And for future improvements, Omnichannel-ERP was exclusively built for linking inventory and in-store point of sale (POS) to Shopee Open API, which is already a limitation. Future developers can innovate the proposed application design by adding a connection feature at the very least to leading online marketplaces, making room for enterprises to expand their market reach. Finally, incorporating Artificial Intelligence (AI) and Machine Learning (ML) into the main data pipeline, where it can be optimized to make predictive analytics and advance automation which are very common to modern companies.

REFERENCES

Akula, R. P. (2024). The Impact of Cloud-Based Enterprise Performance Management on Inventory Optimization and Financial Agility in Omnichannel Retail. *World Journal of Advanced Engineering Technology and Sciences*, 13(1), 1036-1045.

Chen, J., Liang, Y., Shen, H., Shen, Z. J. M., & Xue, M. (2022). Offline-channel planning in smart omnichannel retailing. *Manufacturing & Service Operations Management*, 24(5), 2444-2462.

Febriani, A., Sopha, B. M., & Wibisono, M. A. (2025). Unlocking omnichannel success for small and medium-sized enterprises: A systematic review of technologies, barriers, and enablers. *Journal of Industrial Engineering and Management*, *18*(3), 553-576.

ACADEMIC INTEGRITY DECLARATION

I certify that this report is our original work. All sources have been properly cited, and any AI tools used were limited to idea support and language assistance.

Signature:

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